Appendix S1: Detailed methodology

Extracting demographic data from mapped datasets

To classify genets as survivors or recruits, we used a computer program that tracks genets based on their spatial locations within the quadrats (Lauenroth and Adler 2008). For example, when a genet present in year t+1 overlaps in area with a conspecific genet present in year t, we assume it to be the same genet. On the other hand, if a genet in year t+1 is more than 5 cm from any conspecific genet present in year t, we classify it as a recruit. Our approach recognizes that genets may fragment and/or coalesce over the study period. Some plants were identified by mappers as seedlings; we classified these plants as recruits regardless of their proximity to existing genets.

In a previous survival analysis of Kansas prairie species (Lauenroth and Adler 2008), we only included genets that established after the first observed year in each quadrat. For this study, we included plants present in the first year as well by visually comparing the first and second year's maps from each quadrat and assigning each polygon or cluster of polygons a unique identification (Dalgleish et al. unpublished manuscript). The caespitose growth form of our study grasses, which often results in clusters of closely grouped polygons, gave us confidence in these subjective decisions.

For parameterizing our models, we chose to represent each genet in each year as a circle with an area equal to the sum of all the polygons that compose the genet and a location corresponding to the polygons' centroid (Figure S1). Multiple genets can have virtually identical centroids when the polygons that compose each genet are interspersed. Very small plants were originally mapped as points; we represented those as circles with an area of 0.25 cm^2. The distance between two genets was defined to be the distance between their centroids. Information

on the fate of plants located along the quadrat edges was not used in the statistical modeling of growth and survival. However, these edge plants were included in the calculations of the amount of neighborhood crowding experienced by more centrally located genets.

Statistical modeling of survival and growth

We assume that the survival probability and growth of individual genets is a function of genet size, the neighborhood-scale crowding experienced by the genet, temporal variation among years, and permanent spatial variation among groups of quadrats (the 4-6 quadrats within each group are generally within 50 m of each other, while groups may be separated by up to 3 km).

Our description of neighborhood crowding assumes that the influence of neighbors on a focal individual depends on the distance, *d,* to the neighbor and the neighbor's size, *u*:

$$w_{ijm,t} = \sum_{k} e^{-\alpha_{jm} d^2_{ijkm,t}} u_{km,t} \qquad (1)$$

Here, $w_{ijm,t}$ is the crowding that genet *i* in species *j* in year *t* experiences from neighbors of species *m*, α_{jm} determines the spatial scale over which neighbors of species *m* exert influence on a genet of species *j*, *k* indexes all the focal genet's neighbors of species *m* at time *t,* and $d_{ijkm,t}$ is the distance between genet *i* in species *j* and genet *k* in species *m*. Our use of squared distances implies a Gaussian competition kernel. A negative exponential kernel performed marginally better in the statistical models, but caused simulations of the individual-based model to crash. The total crowding impact on a genet was assumed to be a weighted sum of the impacts from each species, i.e.,

$$\boldsymbol{w}^{V}_{ij,t} = \sum_{m=1}^{4} \omega^{V}_{jm,t} w_{ijm,t} \qquad (2)$$

where *V*=*S* or *G*, indicating Survival or Growth. Note that the competition coefficients ω that

determine the relative impacts of different species depend on *V*, but the distance-weighted neighborhood crowding *w* does not, because the estimated values of α for survival and growth were similar enough (as discussed below) that we assumed a common value.

We modeled the survival probability, *S*, of genet *i* in species *j* and group *g* from time *t* to *t*+1 as

$$Logit(S_{ijg,t}) = \gamma^S_{j,t} + \phi^S_{jg} + \beta^S_{j,t} u_{ij,t} + \boldsymbol{w}^S_{ij,t} \quad (3)$$

where γ is a time-dependent intercept, ϕ is the coefficient for the effect of group, and β is the coefficient that determines how genet size, *u*, influences survival. Fitting this model to the data included estimation of the competition coefficients $\omega^S_{jm,t}$.

Our model for changes in genet size from time *t* to *t*+1, conditional on survival, has a similar structure:

$$u_{ijg,t+1} = \gamma^G_{j,t} + \phi^G_{jg} + \beta^G_{j,t} u_{ij,t} + \boldsymbol{w}^G_{ij,t} + \epsilon^G_{ij,t}. \quad (4)$$

We modeled the variance in growth as a nonlinear function of predicted genet size:

$$Var(\epsilon^G_{ij,t}) = a\, e^{b\, u_{ij,t+1}}. \quad (5)$$

Equations (4) and (5) were fit by a two-stage least squares procedure, first fitting (4) on the assumption of constant error variance by least squares (which is unbiased, but not optimally efficient) and then fitting (5) by nonlinear least-squares regression of the squared residuals on the corresponding predicted mean value. The departures from constant variance were small for all species (the largest was for *A. tripartita*, which had 13% coefficient of variation in the fitted size-specific error standard deviation values) so fitting by least squares entails only a very small loss of efficiency. The near-constancy of the variance also justifies *post hoc* our use in model selection of AIC values from the least-squares fit of the mean, which are computed on the assumption of constant error variance.

We considered three questions about model structure. First, which parameters should vary through time? Equations 3 and 4 allow the intercept, γ, size effect, β, and crowding effects, ω, to vary through time, but the data should determine whether we model these coefficients as constant or time-varying. Second, does size, u, interact with neighborhood crowding, w? Third, in a multispecies model, should α vary with the focal species, the neighbor species, or both?

We addressed these questions by fitting a number of alternative models in a generalized linear models (GLM) framework, treating all time-varying parameters as fixed effects (fitting a separate parameter for each effect for each year) and using AIC to identify the model with the highest predictive accuracy for each species. After completing model selection, we then fit mixed-effects versions of the models. The first step was fitting a series of single-species survival and growth regressions of varying complexity for a range of values of α. We explored dropping terms from Eqs. 3 and 4, adding interactions, and allowing different model parameters to vary through time. All models were implemented in R 2.9 using function "glm" (R Development Core Team 2009). We found that the effects of model complexity and α were largely independent: the selected (i.e., lowest AIC) regression model rarely depended on the value of α, and vice versa (Figure S11). We also found that, for most species, the selected value of α was very similar for survival and growth (Figure S11). Therefore, we decided to use a common value of α for both life stages. When the selected values differed, we chose the value of α that minimized the combined likelihoods of the selected survival and growth regression models as a function of α.

After selecting α_{jj} values for each single-species model, the next step was determining how to translate those values into α_{jm}'s for interspecific interactions. We considered three possibilities. In the "Effects" model, each species has a characteristic zone of influence: an

individual of a given size will affect all its neighbors similarly, regardless of the neighbor's species. For this model, the value of α in Eqn. 1 depends on the species identity of the neighbor, not the focal individual (i.e., $\alpha_{jm}=\alpha_m$, where α_m is the value from the selected single-species model for species *m*) . Conversely, in the "Response" model, the value of α in Eqn. 1 depends only on the species of the focal individual (i.e., $\alpha_{jm}=\alpha_j$). The idea is that some species may only perceive very local interactions, while other species integrate across larger areas. Finally, the "Interaction" model assumes that both effects and responses are relevant. The value of α will therefore be an average of the single-species values for the focal species and the neighbor. To compare these three possibilities, we fit survival and growth regressions for each species using multispecies neighborhoods, with either the Effects, Responses or Interaction models defining the interspecific α's based on the single-species values. At this stage we also used the model structures (interactions and year effects) of the selected the single-species models. We found that the Effects model was superior more often than the other two models, although for most species the differences were small (Table S2). Based on this result we used the Effects model for all species.

The last step was fitting mixed-effects models incorporating year and spatial location as random effects and α values for all species based on the Effects model. We fitted the models using "lmer" (for growth) and “glmer” (for survival) in R package "lme4" (version 0.999375-32). The survival regression included random year effects on the intercept for all four species and on the size effect for all species except *Poa secunda*. We included location (group) random effects for *H. comata* and *Poa secunda* and a size-by-crowding interaction for all three grass species but not for *A. tripartita*. For the growth model, we included random year effects on both the intercept and the size effect for all species and location (group) random effects for all species

expect *Poa secunda.* Interactions between size and crowding caused simulations of the individual-based to crash, so we excluded from consideration growth models that included size-by-crowding interactions.

Statistical modeling of recruitment

In contrast to survival and growth, which we modeled at the individual level, we model recruitment at the quadrat level because we cannot determine which recruits were produced by which potential parent genets. We assume that the number of individuals, y, of species j recruiting at time t+1 in location q follows a negative binomial distribution (the observations appeared overdispersed relative to a Poisson model):

$$y_{jq,t+1} \sim NegBin(\lambda_{jq,t+1}, \theta) \tag{6}$$

where λ is the mean intensity and θ is the size parameter. In turn, λ depends on the composition of the quadrat in the previous year :

$$\lambda_{jq,t+1} = C'_{jq,t} e^{(\gamma_t^R + \phi_g^R + \boldsymbol{\omega}_t^R \boldsymbol{C}'_{qt})} \quad . \tag{7}$$

C'_{jqt} is the "effective cover" (cm^2) of species j in quadrat q at time t, γ is a time-dependent intercept, ϕ is a coefficient for the effect of group location, $\boldsymbol{\omega}$ is a vector of coefficients that determine the strength of intra- and interspecific density-dependence, and $\boldsymbol{C'}$ is the vector of "effective" cover of each species. The model is a form of a Ricker (1954) equation for discrete time population growth. By estimating each species' effective cover in a quadrat, we recognized that plants outside the mapped quadrat may contribute recruits to the focal quadrat, and vice versa. We estimated effective cover as a mixture of the observed cover, C, in the focal quadrat, q, and the mean cover across the group, g, in which the quadrat is located:

$$C'_{jqt} = p_j C_{jqt} + (1 - p_j) \bar{C}_{jgt} \tag{8}$$

where p is the mixing fraction between 0 and 1.

We first fit the recruitment model by maximum likelihood treating year and groups as fixed effects. We compared models with and without year and group effects, and with and without estimation of the latent effective cover ("without" meaning that the value of p was fixed at 1). We then fit a mixed-effects model incorporating as random effects all year and group effects that were significant in the fixed-effects analysis, using WinBUGS 1.4 (Lunn et al. 2000). We used uninformative priors, ran the Markov Chain Monte Carlo simulation for 10,000 interations after a 10,000 iteration burn in period, and checked for convergence of three parameter chains using Brooks and Gelman's (1998) potential scale reduction factor (all values of $\hat{r}$ were <1.1).

Individual-based model

Once the recruitment, growth and survival regressions are fit, implementing a spatially-explicit individual-based model (IBM) is straightforward. First, we initialized the model by randomly drawing sizes for genets of each species from their observed size distributions, until the specified initial cover level was reached. We assigned each individual genet x and y coordinates drawn from a uniform distribution. At each time step of the model we 1) select year-specific parameters from the fitted random effects models, as described below; 2) apply the recruitment regression to determine the number of new recruits at time t+1 based on cover at time t; 3) apply the survival regression to determine which genets survive from time t to t+1, based on neighborhood crowding, and 4) apply the growth regression to determine the sizes of surviving genets at time t+1 based on neighborhood crowding. We then update the simulated landscape to reflect the arrival of new recruits (which are assigned random spatial coordinates) and genet survival and growth. We ran the simulation on a toroidal landscape to avoid edge

effects. The size of the modeled landscape was limited by the computer memory required to calculate the matrix of distances among all genets, especially in rare high-recruitment years.

To simulate the time-varying environment, at each time step we randomly selected one of the 22 sets of fitted, year-specific parameters from the mixed-effects models. This approach is a compromise between the "matrix selection" (fixed effects) and "element selection" (random effects) approaches to parameterizing stochastic matrix or integral projection models. The year-specific parameter sets obtained from the mixed-effects models combine year-specific data with information from the full data set, in order to estimate the actual amount of among-year parameter variability, without the additional variance in fitted parameters due to sampling errors that is present in fixed-effects parameter estimates. However, the fitted year-specific parameters from the mixed model are not strongly influenced by the simplifying assumptions required for fitting the model (e.g., multivariate Gaussian distributions for parameter variation), whereas samples drawn from the fitted mixed model would necessarily satisfy the model's assumptions exactly. To simulate a constant environment, we can either apply one of the fitted year-specific parameters sets at every time step or use the mean of the estimated year-specific random effects.

In addition to environmental stochasticity represented by year-to-year parameter variation, demographic stochasticity arises from all three vital rates. The number of recruits is drawn from a negative binomial distribution, survival of each genet is drawn from a Bernoulli distribution, and changes in genet sizes are drawn from a normal distribution with size-dependent variance (Eq. 5).

We evaluated the performance of the IBM by comparing simulated and observed abundances, genet size distributions, and spatial patterns. We simulated time series of abundances in each quadrat by initializing the IBM with the actual conditions (plant sizes and

locations) observed in the first year the quadrat was censused. We then projected the model forward in time, drawing from the year-specific parameters in chronological order, and applying location-specific parameters as appropriate for each quadrat. For these simulations we used absorbing boundaries. Where data gaps in the observed time series occurred, we re-initialized the IBM with the plant sizes and locations observed in the year following the gap. For each quadrat, we ran 50 replicate simulation runs and averaged cover and density across runs for each species. To simulate data for analysis of size and spatial distributions, we ran 50 simulations of 100 time steps for an average location, and saved plant sizes and spatial coordinates every 10 time steps after a 50 year burn in period. We used Ripley's K (Ripley 1977) and cross-K functions to analyze intra- and interspecific spatial patterns, respectively, using the "spatstat" package for R.

Integral-projection model

While the demographic stochasticity present in the IBM is appropriate for analyzing time to extinction, it complicates analyses of invasion growth rates and equilibrium abundances. Therefore, we built an environmentally stochastic but demographically deterministic version of the model using an integral projection models (IPM) framework. IPMs provide an elegant way to model the dynamics of continuously (size) structured populations in discrete time (Easterling et al. 2000, Ellner and Rees 2006, 2007). However, existing IPMs have either treated single species in a non-spatial context, or focused exclusively on predicting asymptotic rates of spread for an invasive species (Jongejans et al, *in review*). We extended these approaches to accommodate our multispecies, spatially-explicit data using a mean field approximation.

In our IPM, the population of species *j* is represented by a density function $n(u_j, t)$ which gives the density of genets of size *u* at time *t*, with genet size on natural-log scale, i.e.

$n(u_j, t)du$ is the number of genets whose area (on arithmetic scale) is between $\exp(u_j)$ and $\exp(u_j + du)$. The density function for size *v* at time *t*+1 is given by

$$n(v_j, t+1) = \int_{L_j}^{U_j} k_j(v_j, u_j, \bar{\boldsymbol{w}}_j(u_j))\, n(u_j, t) \tag{9}$$

where the kernel $k_j(v_j, u_j, \bar{\boldsymbol{w}}_j)$ describes all possible transitions from size *u* to *v* and $\bar{\boldsymbol{w}}_j$ is a vector whose elements are the average crowding experienced by an individual of size u_j in species *j* from each of the other species. We describe below how $\bar{\boldsymbol{w}}_j$ is calculated from the density functions for the species in the model. The integral is evaluated over all possible sizes from a lower size limit *L* to an upper size limit *U* that extends past the range of observed sizes.

The kernel is constructed from the fitted survival (*S*), growth (*G*), and recruitment (*R*) models:

$$k_j(v_j, u_j, \bar{\boldsymbol{w}}_j) = S_j(u_j, \bar{\boldsymbol{w}}_j(u_j))\, G_j(v_j, u_j, \bar{\boldsymbol{w}}_j(u_j)) + R_j(v_j, u_j, \bar{\boldsymbol{w}}_j) \tag{10}$$

S is given by Eq. (3) and *G* by Eq. (4). The recruitment function, *R*, defined in Eq. 6 and 7, gives the number of new recruits produced per quadrat. To incorporate this recruitment function into the IPM, we assumed that fecundity increases linearly with size, $R_j(v_j, u_j, \bar{\boldsymbol{w}}_j) = e^{u_j} R_j(v_j, \bar{\boldsymbol{w}}_j)$ (Dalgleish et al. *unpublished manuscript*); this has the consequence that recruitment by any species in the IPM is proportional to the species' total cover, as desired.

In the vital rate regressions and the IBM, we calculate a neighborhood crowding w_{ij} unique to each individual *i* based on the spatial locations and sizes of neighboring plants (Eq. 1). This spatially-explicit approach cannot be extended to the IPM. Therefore, we developed spatially-implicit approximations to $\bar{\boldsymbol{w}}_j$ that incorporate the essential features of spatially-explicit neighborhood competition. The simplest mean-field approximation assumes that plant locations (the centers of the circles representing individual genets) are distributed randomly and

independently. In this approximation (see next section, *Mean-field approximations for neighborhood competition*), the mean crowding exerted by species *k* on species *j* is given by

$$\bar{w}_{jk} = \frac{\pi N_k \bar{X}_k}{\alpha_k A} \tag{11}$$

where *N* is the average density of species *k* (individuals per quadrat), $\bar{X}$ is the average size of species *k* individuals (on absolute scale), α is the spatial scale over which species k affects neighbors (defined in Eq. 1), and *A* is the area of a quadrat, in the same units as $\bar{X}$.

IPM simulations using equation (11) generated equilibrium abundances that were unrealistically low, indicating that this approximation overestimated the strength of neighborhood crowding. The explanation for this overestimate is non-random spatial distribution of genets: if plants are distributed more regularly than expected by chance, average crowding will be reduced. Indeed, when we analyzed the spatial point patterns of conspecifics in the observed data and in IBM simulations, we found that while very small individuals were distributed randomly, large genets had a regular distribution (Figure 2, Figure S2). Thus, large plants experience less conspecific crowding than small plants on average. This pattern is much weaker for heterospecific spatial patterns.

The principal feature of the overdispersion of large plants is that they do not overlap. More specifically, large plants have very few conspecific neighbors closer than twice the mean radius of large plants of their species (Figure 2, Figure S6). We therefore modified our mean-field approximation by assuming that plants are distributed at random subject to a "no overlap" rule which requires that conspecific genets cannot occupy overlapping areas (see *Mean-field approximations for neighborhood competition*). With a no-overlap constraint, the mean conspecific crowding experienced by a species *j* individual of radius *r* due to neighbors of species *k* is given by

$$\bar{w}_{jk}(r) = 2\pi \int_{r}^{\infty} z\, e^{-\alpha_{jk} z^2} C_k(z-r)\, dz \qquad (12)$$

where $C_k(z-r)$ is the total cover of plants of species *k* of radius $z-r$ or smaller. When we simulated the IPM using Eq. (12) to estimate $\bar{w}$ for $k=j$ and Eq. (11) for $k \neq j$, the model generated realistic abundances.

The IPM and IBM thus represent two different approaches to modeling the effect of spatial dynamics on interspecific competition. In the IBM, the spatial patterns and dynamics are emergent properties of the assumed individual-level interaction rules and the fitted neighborhood competition parameters. The IPM lets us "hard wire" into the model important features of the empirically observed spatial patterns. The IBM therefore has the advantage that it can be used to study the feedbacks between individual-level properties and spatial dynamics (e.g., how would spatial patterns change under altered environmental variability, or local extinction of one of the species?), and the disadvantage that errors in predicted spatial patterns may have consequences for the predictions of average abundance that are the focus of this article.

Mean-field approximations for neighborhood competition

Here we derive the mean-field approximations for neighborhood competitive pressure that are used in the IPMs. In the statistical models for survival and growth, the impact of neighbor competition on a focal genet *i* in species *j,* due to neighbors of species *m,* is (omitting for clarity the time index *t*)

$$w_{ijm} = \sum_{k} e^{-\alpha_{jm} d_{ijkm}^2} u_{km} \quad . \qquad (13)$$

Our goal here is to derive approximations to the mean impact, $\bar{w}_{jm}$, of species *m* neighbors on a genet in species *j*.

Our first approximation assumes that genet locations (the centers of the circles representing individual genets) are distributed randomly and independently, without any regard for the locations of other genets, over a spatially infinite homogeneous region. Consider an infinitesimal annulus around a focal genet in species *j*, consisting of the region between the circles of radius *z* and *z*+*dz* centered on the genet. The total area of the annulus is $2\pi z\, dz$, so the expected contribution of the annulus to the impact on the focal genet is $e^{-\alpha_{jm} z^2} 2\pi z\, dz$ multiplied by the expected total cover C_m of the competitor species; here "total cover" means the total area of plants as a fraction of the total ground area. Therefore

$$\bar{w}_{jm} = \int_0^\infty C_m e^{-\alpha_{jm} z^2} 2\pi z\, dz = \frac{\pi C_m}{\alpha_{jm}} \tag{14}$$

We can express C_m as $N_m \frac{\bar{X}_m}{A}$, where N_m is the total number of competitor genets per quadrat, $\bar{X}_m$ is the average area of competitor genets, and *A* is the area of a quadrat. This gives the formula for $\bar{w}_{jm}$ in Equation 11.

The density function n the IPM, $n_m(u,t)$, gives the density (number of individuals per quadrat) of genets of size *u* at time *t*, with genet size on natural-log scale. The total cover of species *m* in year *t* can therefore be calculated as

$$C_m = \frac{1}{A} \int_{L_m}^{U_m} e^u n_m(u,t)\, du \tag{15}$$

where L_m and U_m are the minimum and maximum sizes (on log scale) of species *m* genets. Our computer code for the IPM uses the midpoint rule to evaluate integrals numerically, as in Ellner and Rees (2007). The density function is represented by a vector **n** consisting of approximate function values at the vector of mesh points **v,** so the numerical approximation for C_m is **h*sum(n*exp(v))** where *h* is the mesh size (distance between mesh points).

Our second approximation takes into account the main feature of the overdispersion of large plants, namely that conspecifics do not overlap, and apart from this restriction it is still assumed that individuals are distributed at random in space. Our "no niche" models assume that there is also no overlap possible between genets of different species. Consider a focal genet of species *j* of radius *r* , and a species *m* (which may be *j*) that cannot overlap with species *j*. The "no- overlap" zone around the focal genet is negligibly small compared to the total area of the habitat, so the existence of the no-habitat zone has a negligible effect on the density of potential competitors in locations that they can occupy. Let $C_m(\rho)$ denote the total cover of species *m* genets of radius ρ or less. Individuals of species *m* can have their centers in the annulus at radius *z* if their radius is $z - r$ or less. The competitive pressure experienced by the focal plant from neighbors of species *m* is then

$$\bar{w}_{jm}(r) = 2\pi \int_r^\infty z\, e^{-\alpha_{jm} z^2} C_m(z-r)\, dz \quad . \tag{16}$$

To avoid numerical evaluation of the infinite integral, note that if ρ greater than or equal to radius r_U corresponding to the upper size limit for species m, then $C_m(\rho)$ equals the total cover for species m, C_m . Therefore

$$\begin{aligned} w_{jm} &= 2\pi \left(\int_r^{r+r_U} z\, e^{-\alpha_{jm} z^2} C_m(z-r)\, dz + C_m \int_{r+r_U}^\infty z\, e^{-\alpha_{jm} z^2} dz \right) \\ &= 2\pi \int_r^{r+r_U} z\, e^{-\alpha_{jm} z^2} C_m(z-r)\, dz + \frac{\pi C_m e^{-\alpha (r+r_U)^2}}{\alpha_{jm}} \end{aligned} \tag{17}$$

A more refined version of this approximation could take into account the effect of no-overlap zones on the mean abundance of species in locations that they can occupy, but in this paper we used the simplest version of the no-overlap approximation.

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